

Warmup: data wrangling

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Baseball data

The `teams` dataset contains information on wins, losses, and attendance for three MLB teams (Anaheim, Baltimore, and the New York Mets) between 2021 and 2023:

`teams`

	yearID	franchID	W	L	attendance
1	2021	BAL	52	110	793229
2	2021	ANA	77	85	1515689
3	2021	NYM	77	85	1511926
4	2022	BAL	83	79	1368367
5	2022	ANA	73	89	2457461
6	2022	NYM	101	61	2564737
7	2023	BAL	101	61	1936798
8	2023	ANA	73	89	2640575
9	2023	NYM	75	87	2573555

Questions

1. The following chunk of code is run, producing the output below. What is the `select` function doing?

```
teams |>
  select(yearID, franchID, W, L)
```

	yearID	franchID	W	L
1	2021	BAL	52	110
2	2021	ANA	77	85
3	2021	NYM	77	85
4	2022	BAL	83	79
5	2022	ANA	73	89
6	2022	NYM	101	61
7	2023	BAL	101	61
8	2023	ANA	73	89
9	2023	NYM	75	87

2. After running the following code, what will be the dimensions (number of rows and columns) of the `team_sub` dataset? Of the `teams` dataset?

```
team_sub <- teams |>
  filter(franchID %in% c("BAL", "NYM"))
```

3. After running the following code, what will be the dimensions (number of rows and columns) of the `team_stats` dataset?

```
team_stats <- teams |>
  group_by(franchID) |>
  summarize(max_w = max(W), min_w = min(W))
```

4. Give the output of the following code:

```
teams |>
  filter(franchID %in% c("BAL", "NYM")) |>
  group_by(franchID) |>
  summarize(max_attend = max(attendance))
```